

# C Programming Practice Problems

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From Variable Declaration to Conditional Statements

## Section 1: Variable Declarations, Data Types, and I/O (Problems 1–5)

### Problem 1: Personal Information Card

**Topic: Variable Declaration, printf()**

Write a C program that declares variables to store your name (string), age (integer), and height in meters (float). Print these values in a formatted manner using printf().

**Code:**

```
#include <stdio.h>
int main() {
    char name[] = "Alice";
    int age = 20;
    float height = 1.65;
    printf("Name: %s\n", name);
    printf("Age: %d years\n", age);
    printf("Height: %.2f meters\n", height);
    return 0;
}
```

**Output**

| Output              |
|---------------------|
| Name: Alice         |
| Age: 20 years       |
| Height: 1.65 meters |

---

## Problem 2: Shopping Bill Calculator

### Topic: Arithmetic Operations, Variable Initialization

Declare variables for the price of three items. Calculate the total cost and the average price. Display all values with 2 decimal places.

#### Code:

```
#include <stdio.h>

int main() {
    float item1 = 45.50, item2 = 120.00, item3 = 34.99;
    float total, average;
    total = item1 + item2 + item3;
    average = total / 3.0;
    printf("Item 1: %.2f\n", item1);
    printf("Item 2: %.2f\n", item2);
    printf("Item 3: %.2f\n", item3);
    printf("Total Bill: %.2f\n", total);
    printf("Average Price: %.2f\n", average);
    return 0;
}
```

#### Output

##### Output

```
Item 1: 45.50
Item 2: 120.00
Item 3: 34.99
Total Bill: 200.49
Average Price: 66.83
```

---

### Problem 3: Circle Metrics

#### Topic: Constants, scanf(), printf(), Floating-Point Arithmetic

Write a program that reads the radius of a circle from the user. Declare a constant for PI. Calculate and print the diameter, circumference, and area.

#### Code:

```
#include <stdio.h>
#define PI 3.14159
int main() {
    float radius, diameter, circumference, area;
    printf("Enter the radius of the circle: ");
    scanf("%f", &radius);
    diameter = 2 * radius;
    circumference = 2 * PI * radius;
    area = PI * radius * radius;
    printf("Diameter: %.2f\n", diameter);
    printf("Circumference: %.2f\n", circumference);
    printf("Area: %.2f\n", area);
    return 0;
}
```

#### Output

##### Output

```
Enter the radius of the circle: 28
Diameter: 56.00
Circumference: 175.93
Area: 2463.01
```

---

## Problem 4: Temperature Converter

### Topic: Variable Declaration, Assignment, Expression Evaluation

Declare a variable celsius, initialize it to 37.5, convert to Fahrenheit. Then read a new temperature from user and convert back.

#### Code:

```
#include <stdio.h>
int main() {
    float celsius = 37.5, fahrenheit;
    fahrenheit = (celsius * 9.0 / 5.0) + 32;
    printf("%.2f C = %.2f F\n", celsius, fahrenheit);
    printf("Enter temperature in Fahrenheit: ");
    scanf("%f", &fahrenheit);
    celsius = (fahrenheit - 32) * 5.0 / 9.0;
    printf("%.2f F = %.2f C\n", fahrenheit, celsius);
    return 0;
}
```

#### Output

##### Output

```
37.50 C = 99.50 F
Enter temperature in Fahrenheit: 273
273.00 F = 133.89 C
```

## Problem 5: Student Marks Summary

### Topic: Integer vs. Float Division, Multiple Inputs

Read five subject marks (integers) for a student. Calculate total, average, and percentage (max 100 per subject).

#### Code:

```
#include <stdio.h>

int main() {
    int m1, m2, m3, m4, m5, total;
    float average, percentage;
    printf("Enter marks for 5 subjects (0-100 each): ");
    scanf("%d %d %d %d %d", &m1, &m2, &m3, &m4, &m5);
    total = m1 + m2 + m3 + m4 + m5;
    average = total / 5.0;
    percentage = (total / 500.0) * 100;
    printf("Total Marks: %d\n", total);
    printf("Average Marks: %.2f\n", average);
    printf("Percentage: %.2f%%\n", percentage);
    return 0;
}
```

#### Output

```
Output
Enter marks for 5 subjects (0-100 each): 86
96
75
75
60
Total Marks: 392
Average Marks: 78.40
Percentage: 78.40%
```

---

## Section 2: Operators and Expressions (Problems 6–10)

### Problem 6: Swap Two Numbers

**Topic: Assignment Operators, Temporary Variables**

Declare two integer variables, swap their contents using a temporary variable. Print before and after. Extension: swap without temp variable.

#### Code:

```
#include <stdio.h>

int main() {
    int a = 10, b = 20, temp;
    printf("Before swap: a = %d, b = %d\n", a, b);
    temp = a;
    a = b;
    b = temp;
    printf("After swap (with temp): a = %d, b = %d\n", a, b);
    /* Extension: Without temp (arithmetic - risk of overflow) */
    a = a + b;
    b = a - b;
    a = a - b;
    printf("After swap (without temp): a = %d, b = %d\n", a, b);
    return 0;
}
```

#### Output

##### Output

```
Before swap: a = 10, b = 20
After swap (with temp): a = 20, b = 10
After swap (without temp): a = 10, b = 20
```

---

## Problem 7: Time Decomposition

### Topic: Modulo and Division Operators

Read total seconds from user. Decompose into hours, minutes, and remaining seconds.

#### Code:

```
#include <stdio.h>
int main() {
    int total_seconds, hours, minutes, seconds;
    printf("Enter total seconds: ");
    scanf("%d", &total_seconds);
    hours = total_seconds / 3600;
    minutes = (total_seconds % 3600) / 60;
    seconds = total_seconds % 60;
    printf("%d seconds = %d hours, %d minutes, %d seconds\n",
           total_seconds, hours, minutes, seconds);
    return 0;
}
```

#### Output

##### Output

```
Enter total seconds: 445
445 seconds = 0 hours, 7 minutes, 25 seconds
```

---

## Problem 8: Digit Extractor

### Topic: Arithmetic Operators, Integer Division

Read a 4-digit positive integer. Extract and print each digit using only division and modulo.

#### Code:

```
#include <stdio.h>
int main() {
    int num;
    printf("Enter a 4-digit number: ");
    scanf("%d", &num);
    printf("Thousands: %d\n", num / 1000);
    printf("Hundreds: %d\n", (num / 100) % 10);
    printf("Tens: %d\n", (num / 10) % 10);
    printf("Units: %d\n", num % 10);
    return 0;
}
```

#### Output

| Output                       |
|------------------------------|
| Enter a 4-digit number: 1228 |
| Thousands: 1                 |
| Hundreds: 2                  |
| Tens: 2                      |
| Units: 8                     |



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## Problem 9: Compound Interest Calculator

**Topic: Type Casting, pow() Function, <math.h>**

Read principal, rate, and time. Calculate compound interest using pow() from math.h.

### Code:

```
#include <stdio.h>
#include <math.h>
int main() {
    double principal, rate, time, amount, ci;
    printf("Enter principal amount: ");
    scanf("%lf", &principal);
    printf("Enter rate of interest: ");
    scanf("%lf", &rate);
    printf("Enter time in years: ");
    scanf("%lf", &time);
    amount = principal * pow(1 + rate / 100, time);
    ci = amount - principal;
    printf("Amount: %.2f\n", amount);
    printf("Compound Interest: %.2f\n", ci);
    return 0;
}
```

### Output

| Output                        |
|-------------------------------|
| Enter principal amount: 1000  |
| Enter rate of interest: 2000  |
| Enter time in years: 3        |
| Amount: 9261000.00            |
| Compound Interest: 9260000.00 |

---

## Problem 10: Relational and Logical Operators Explorer

### Topic: Relational Operators, Logical Operators, Truth Tables

Declare a=5, b=10, c=5. Evaluate and print truth values of given expressions.

#### Code:

```
#include <stdio.h>
int main() {
    int a = 5, b = 10, c = 5;
    printf("a == c: %d\n", a == c);
    printf("a != b: %d\n", a != b);
    printf("a > b: %d\n", a > b);
    printf("b >= c: %d\n", b >= c);
    printf("(a == c) && (b > a): %d\n", (a == c) && (b > a));
    printf("(a > b) || (c < b): %d\n", (a > b) || (c < b));
    printf("!(a == c): %d\n", !(a == c));
    printf("(a < b) && (b < c) || (a == c): %d\n",
           (a < b) && (b < c) || (a == c));
    return 0;
}
```

#### Output

| Output                            |
|-----------------------------------|
| a == c: 1                         |
| a != b: 1                         |
| a > b: 0                          |
| b >= c: 1                         |
| (a == c) && (b > a): 1            |
| (a > b)    (c < b): 1             |
| !(a == c): 0                      |
| (a < b) && (b < c)    (a == c): 1 |

---

## Section 3: Simple if and if-else Statements (Problems 11–15)

### Problem 11: Even or Odd

**Topic: if-else, Modulo Operator**

Read an integer. Check whether it is even or odd using if-else.

#### Code:

```
#include <stdio.h>
int main() {
    int num;
    printf("Enter an integer: ");
    scanf("%d", &num);
    if (num % 2 == 0) {
        printf("%d is even.\n", num);
    } else {
        printf("%d is odd.\n", num);
    }
    return 0;
}
```

#### Output

##### Output

```
Enter an integer: 18
18 is even.
```

---

## Problem 12: Vote Eligibility Checker

### Topic: Simple if

Read a person's age. If 18 or above, print eligible. Extension: check for negative age.

### Code:

```
#include <stdio.h>

int main() {
    int age;
    printf("Enter your age: ");
    scanf("%d", &age);

    if (age < 0) {
        printf("Invalid age!\n");
    }
    if (age >= 18) {
        printf("You are eligible to vote.\n");
    }
    return 0;
}
```

### Output

#### Output

```
Enter your age: 19
You are eligible to vote.
```

---

### Problem 13: Maximum of Two Numbers

#### Topic: if-else Comparison

Read two integers. Use if-else to determine which is larger, or if equal.

#### Code:

```
#include <stdio.h>
int main() {
    int a, b;
    printf("Enter two integers: ");
    scanf("%d %d", &a, &b);
    if (a > b) {
        printf("%d is larger than %d.\n", a, b);
    } else if (b > a) {
        printf("%d is larger than %d.\n", b, a);
    } else {
        printf("Both numbers are equal.\n");
    }
    return 0;
}
```

#### Output

##### Output

```
Enter two integers: 12 28
28 is larger than 12.
```

---

### Problem 14: Pass or Fail

#### Topic: if-else, Threshold Checking

Read marks (0-100). Print Pass if  $\geq 40$ , Fail otherwise. Add congratulatory message if  $\geq 90$ .

#### Code:

```
#include <stdio.h>

int main() {
    int marks;
    printf("Enter marks (0-100): ");
    scanf("%d", &marks);
    if (marks >= 40) {
        printf("Pass\n");
        if (marks >= 90) {
            printf("Congratulations! Excellent performance!\n");
        }
    } else {
        printf("Fail\n");
    }
    return 0;
}
```

#### Output

| Output                          |
|---------------------------------|
| Enter marks (0-100): 75<br>Pass |

---

## Problem 15: Character Type Checker

**Topic:** if-else, ASCII Values, Character Handling

Read a single character. Determine if uppercase, lowercase, digit, or special character using ASCII ranges.

### Code:

```
#include <stdio.h>
int main() {
    char ch;
    printf("Enter a character: ");
    scanf(" %c", &ch);
    if (ch >= 'A' && ch <= 'Z') {
        printf("'%c' is an uppercase letter.\n", ch);
    } else if (ch >= 'a' && ch <= 'z') {
        printf("'%c' is a lowercase letter.\n", ch);
    } else if (ch >= '0' && ch <= '9') {
        printf("'%c' is a digit.\n", ch);
    } else {
        printf("'%c' is a special character.\n", ch);
    }
    return 0;
}
```

### Output

#### Output

```
Enter a character: Raja
'R' is an uppercase letter.
```

---

## Section 4: else-if Ladder and Nested Conditionals (Problems 16–20)

### Problem 16: Grade Calculator

**Topic: else-if Ladder**

Read percentage (0-100). Use else-if ladder to assign grade. Handle invalid input.

**Code:**

```
#include <stdio.h>
int main() {
    float percentage;
    printf("Enter percentage (0-100): ");
    scanf("%f", &percentage);
    if (percentage < 0 || percentage > 100) {
        printf("Invalid Input.\n");
    } else if (percentage >= 90) {
        printf("Grade: A\n");
    } else if (percentage >= 80) {
        printf("Grade: B\n");
    } else if (percentage >= 70) {
        printf("Grade: C\n");
    } else if (percentage >= 60) {
        printf("Grade: D\n");
    } else if (percentage >= 50) {
        printf("Grade: E\n");
    } else {
        printf("Grade: F\n");
    }
    return 0;
}
```



## Output

### Output

Enter percentage (0-100): 96.2  
Grade: A

---

## Problem 17: Triangle Validity and Type

### Topic: Nested if, Logical Operators

Read three sides. Check triangle inequality. If valid, determine if equilateral, isosceles, or scalene.

### Code:

```
#include <stdio.h>
int main() {
    int a, b, c;
    printf("Enter three sides of triangle: ");
    scanf("%d %d %d", &a, &b, &c);
    if (a + b > c && a + c > b && b + c > a) {
        printf("Valid triangle.\n");
        if (a == b && b == c) {
            printf("It is an equilateral triangle.\n");
        } else if (a == b || b == c || a == c) {
            printf("It is an isosceles triangle.\n");
        } else {
            printf("It is a scalene triangle.\n");
        }
    } else {
        printf("Invalid triangle.\n");
    }
    return 0;
}
```

## Output

### Output

```
Enter three sides of triangle: 10 20 30
Invalid triangle.
```

---

## Problem 18: Leap Year Checker

### Topic: Nested if-else, Multiple Conditions

Read a year. Determine if leap year using nested if statements with intermediate reasoning.

### Code:

```
#include <stdio.h>
int main() {
    int year;
    printf("Enter a year: ");
    scanf("%d", &year);
    if (year % 4 == 0) {
        if (year % 100 == 0) {
            if (year % 400 == 0) {
                printf("%d is a leap year.\n", year);
            } else {
                printf("%d is not a leap year.\n", year);
            }
        } else {
            printf("%d is a leap year (divisible by 4 but not by 100).\n", year);
        }
    } else {
        printf("%d is not a leap year.\n", year);
    }
    return 0;
}
```

## Output

### Output

```
Enter a year: 2027
2027 is not a leap year.
```

---

## Problem 19: Number Sign and Magnitude

### Topic: Nested if-else

Read an integer. Determine sign and magnitude (single/double/triple digit) using nested if.

### Code:

```
#include <stdio.h>
int main() {
    int num;
    printf("Enter an integer: ");
    scanf("%d", &num);
    if (num > 0) {
        printf("Positive ");
        if (num < 10) {
            printf("single-digit number.\n");
        } else if (num < 100) {
            printf("double-digit number.\n");
        } else {
            printf("triple-digit or more number.\n");
        }
    } else if (num < 0) {
        printf("Negative ");
        if (num > -10) {
            printf("single-digit number.\n");
        } else if (num > -100) {
            printf("double-digit number.\n");
        } else {
            printf("triple-digit or more number.\n");
        }
    } else {
        printf("Zero\n");
    }
    return 0;
}
```

## Output

### Output

```
Enter an integer: 1228
Positive triple-digit or more number.
```

---

## Problem 20: Quadrant Identifier

### Topic: else-if with Logical AND

Read coordinates (x, y). Determine quadrant or axis location using else-if.

### Code:

```
#include <stdio.h>

int main() {
    float x, y;
    printf("Enter coordinates (x y): ");
    scanf("%f %f", &x, &y);
    if (x == 0 && y == 0) {
        printf("The point is at the origin.\n");
    } else if (x == 0) {
        printf("The point lies on the Y-axis.\n");
    } else if (y == 0) {
        printf("The point lies on the X-axis.\n");
    } else if (x > 0 && y > 0) {
        printf("Quadrant I\n");
    } else if (x < 0 && y > 0) {
        printf("Quadrant II\n");
    } else if (x < 0 && y < 0) {
        printf("Quadrant III\n");
    } else {
        printf("Quadrant IV\n");
    }
    return 0;
}
```

## Output

### Output

```
Enter coordinates (x y): 12 28  
Quadrant I
```

---

## Section 5: switch Statement and Ternary Operator (Problems 21–25)

### Problem 21: Simple Calculator using switch

**Topic: switch Statement, break, Arithmetic Operators**

Read two numbers and an operator. Use switch to perform operation. Handle division by zero and invalid operators.

## Code:

```
#include <stdio.h>
int main() {
    float a, b, result;
    char op;
    printf("Enter two numbers: ");
    scanf("%f %f", &a, &b);
    printf("Enter operator (+, -, *, /, %%): ");
    scanf(" %c", &op);
    switch (op) {
        case '+':
            result = a + b;
            printf("Result: %.2f\n", result);
            break;
        case '-':
            result = a - b;
            printf("Result: %.2f\n", result);
            break;
        case '*':
            result = a * b;
            printf("Result: %.2f\n", result);
            break;
        case '/':
            if (b == 0) {
                printf("Error: Division by zero!\n");
            } else {
                result = a / b;
                printf("Result: %.2f\n", result);
            }
    }
}
```

## Output

### Output

```
Enter two numbers: 28
12
Enter operator (+, -, *, /, %%): +
Result: 40.00
```

---

## Problem 22: Day of the Week

### Topic: switch with Integer Input

Read an integer (1-7). Print corresponding day using switch. Handle invalid inputs with default.

#### Code:

```
#include <stdio.h>

int main() {
    int day;
    printf("Enter day number (1-7): ");
    scanf("%d", &day);
    switch (day) {
        case 1:
            printf("Monday\n");
            break;
        case 2:
            printf("Tuesday\n");
            break;
        case 3:
            printf("Wednesday\n");
            break;
        case 4:
            printf("Thursday\n");
            break;
        case 5:
            printf("Friday\n");
            break;
        case 6:
            printf("Saturday (Weekend)\n");
            break;
        case 7:
            printf("Sunday (Weekend)\n");
            break;
        default:
```

## Output

| Output                                 |
|----------------------------------------|
| Enter day number (1-7): 3<br>Wednesday |

---

## Problem 23: Vowel or Consonant using switch

### Topic: switch with Character Input, Fall-through Cases

Read a lowercase letter. Use switch with intentional fall-through to group vowels together.

### Code:

```
#include <stdio.h>
int main() {
    char ch;
    printf("Enter a lowercase letter: ");
    scanf(" %c", &ch);
    switch (ch) {
        case 'a':
        case 'e':
        case 'i':
        case 'o':
        case 'u':
            printf("'%c' is a vowel.\n", ch);
            break;
        default:
            if (ch >= 'a' && ch <= 'z') {
                printf("'%c' is a consonant.\n", ch);
            } else {
                printf("Invalid input.\n");
            }
    }
    return 0;
}
```



## Output

### Output

```
Enter a lowercase letter: h
'h' is a consonant.
```

---

## Problem 24: Ternary Operator Challenges

### Topic: Conditional (Ternary) Operator ? :

Solve four sub-problems using ONLY the ternary operator (no if statements).

### Code:

```
#include <stdio.h>
int main() {
    int a, b, c;
    int num;
    /* 1. Even or Odd */
    printf("Enter a number: ");
    scanf("%d", &num);
    printf("%d is %s\n\n", num, (num % 2 == 0) ? "Even" : "Odd");
    /* 2. Maximum of two numbers */
    printf("Enter two numbers: ");
    scanf("%d %d", &a, &b);
    printf("Larger: %d\n\n", (a > b) ? a : b);
    /* 3. Maximum of three numbers (nested ternary) */
    printf("Enter three numbers: ");
    scanf("%d %d %d", &a, &b, &c);
    int max = (a > b) ? ((a > c) ? a : c) : ((b > c) ? b : c);
    printf("Largest: %d\n\n", max);
    /* 4. Positive, Negative, or Zero (double-nested) */
    printf("Enter a number: ");
    scanf("%d", &num);
    printf("%d is %s\n", num,
        (num > 0) ? "Positive" : (num < 0) ? "Negative" : "Zero");
    return 0;
}
```

## Output

### Output

Enter a number: 45

45 is Odd

Enter two numbers: 12 28

Larger: 28

Enter three numbers: 12 29 09

Largest: 29

Enter a number: 07

7 is Positive

---

## Problem 25: Menu-Driven Number Analyzer

**Topic: switch, if-else, All Concepts Combined**

Write a menu-driven program with options to check even/odd, sign, leap year, max of three, or exit. Uses switch to route and if-else inside cases.

## Code:

```
#include <stdio.h>
int main() {
    int choice, num, a, b, c, year;
    do {
        printf("\n===== MENU =====\n");
        printf("1. Check Even or Odd\n");
        printf("2. Check Positive, Negative, or Zero\n");
        printf("3. Check Leap Year\n");
        printf("4. Find Maximum of Three Numbers\n");
        printf("5. Exit\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);
        switch (choice) {
            case 1:
                printf("Enter a number: ");
                scanf("%d", &num);
                if (num % 2 == 0)
                    printf("%d is even.\n", num);
                else
                    printf("%d is odd.\n", num);
                break;
            case 2:
                printf("Enter a number: ");
                scanf("%d", &num);
                if (num > 0)
                    printf("Positive\n");
                else if (num < 0)
                    printf("Negative\n");
```

## Output

### Output

```
===== MENU =====
1. Check Even or Odd
2. Check Positive, Negative, or Zero
3. Check Leap Year
4. Find Maximum of Three Numbers
5. Exit
Enter your choice: 4
Enter three numbers: 12 28 04
28 is the largest.

===== MENU =====
1. Check Even or Odd
2. Check Positive, Negative, or Zero
3. Check Leap Year
4. Find Maximum of Three Numbers
5. Exit
Enter your choice: 5
Exiting program. Goodbye!
```

---

## Appendix: Recommended Progression

| Problem Range | Concepts Introduced                          | Estimated Time |
|---------------|----------------------------------------------|----------------|
| 1–5           | Variables, I/O, constants, arithmetic        | 1–2 hours      |
| 6–10          | Operators, modulo, math library, logical ops | 2–3 hours      |
| 11–15         | if, if-else, basic branching                 | 2 hours        |
| 16–20         | else-if, nested if, complex conditions       | 2–3 hours      |
| 21–25         | switch, ternary operator, consolidation      | 2–3 hours      |

**Happy Coding!**